Amos Murmu

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Professional Experience

Frontend Developer, Techfanatic Private LLP *July 2025 – Apr 2026*

Kidsaarthi Project *Nov 2025 – Apr 2026*

- Architected a modular SPA using **lazy-loaded routes**, improving load performance across core modules
- Integrated **REST APIs** with authentication and session persistence, ensuring consistent data flow using **RTK Query** and **Axios**
- Designed cache invalidation strategy with RTK Query tag types, reducing redundant API calls and maintaining state consistency across views
- Built reusable components and production-grade forms using **React Hook Form + Zod** across multiple workflows
- Delivered a **PWA**-enabled application using Vite + Workbox, enabling offline access and faster repeat visits
- **Tech:** React, TypeScript, Redux Toolkit, RTK Query, Axios, TailwindCSS, Vite PWA

SiteHub Suite v2 *July 2025 – Nov 2025*

- Developed core **CRUD modules** supporting multi-entity workflows with validation and error handling
- Integrated REST APIs to manage authentication, session handling, and dynamic data flows across modules
- Designed reusable components and centralized state using **React Context API**, reducing duplication across features
- Implemented **Azure Maps** for geocoding and address suggestions, improving data accuracy and user experience
- Integrated **Stripe** checkout flows for subscription management with robust success/error handling
- Delivered production-ready features in an Agile environment; project discontinued due to business decisions
- **Tech:** React, TypeScript, React Hook Form, Zod, Azure Maps, Stripe API

Projects

Fridge: Distributed Cache (Redis-inspired) *GitHub*

- Built a distributed in-memory cache with a **router and three cache nodes** over **TCP sockets**, using a custom binary protocol with message framing for `GET`, `SET`, `DEL`, and `MGET`
- Implemented **consistent hashing** with virtual nodes to distribute keys across nodes, and fan-out of multi-key reads with result aggregation
- Added **TTL expiration**, **sampled-LRU eviction**, and a router-level **L1 cache** to reduce repeated network requests
- Implemented **append-only file (AOF) persistence** with log replay to restore each node's state after restart
- Containerized the cluster with **Docker Compose**; local benchmark of 10K requests at 50 concurrency reached ∼37K req/s with 4.7 ms P99 latency on hot-key reads
- **Tech:** TypeScript, Node.js, TCP, Docker, Docker Compose

Image Processing Service API *GitHub*

- Developed a backend API to process images via **CSV uploads** with request tracking
- Implemented background job processing using **BullMQ + Redis** to handle asynchronous image workflows
- Optimized performance using **Redis caching** to reduce repeated requests and improve response time
- **Tech:** Node.js, Express, PostgreSQL, BullMQ, Redis

Open Source Contributions *Mar 2025 – May 2025*

- **Directus** (Fork): Added extension lifecycle events to `ExtensionManager` and updated documentation (PR #25181, PR #336); fixed a permission bug enabling unauthorized file movements (PR #25187)
- **MotionCanvas** (Fork): Developed `centerArrow` prop for Curve subclasses (PR #1186); improved documentation and documented internal architecture for contributors (PR #1185)

Education

AMPICS, Ganpat University, Mehsana, Gujarat *2023*

BCA *CGPA: 7.6*

Technical Skills

Languages: TypeScript, JavaScript, C++, C, SQL, Bash
Frontend: HTML, CSS, React, TailwindCSS, Redux Toolkit, RTK Query, TanStack
Backend and APIs: Node.js, Express, REST APIs, PostgreSQL, MongoDB, MySQL, Postman
DevOps and Tools: Docker, Docker Compose, GitHub Actions, Arch Linux, Debian, Vim, NeoVim
Core CS: Data Structures and Algorithms, System Design, Distributed Systems, Caching, OOP, Debugging

Achievements

- Medal of Distinction, National Cyber Olympiad (NCO), 2019